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Classification of Internet Security Attacks

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ABSTRACT

In the booming era of interconnected computers, the process of transaction has become a common part of each human's life. It is therefore essential to impose the internet security in order to achieve the privacy from intruders and malicious users. Security is considered as an integral part of internet browsing. It is one of the most important quality attribute in the field of networking. Due to the gradually increasing number of vulnerabilities, the identification of attack is essential. Internet attacks thus must be defined in order to measure security.

KEYWORDS

Internet security, Sniffing, DoS, Replay Attack, SQL Injection Attack.

1. INTRODUCTION

Internet is increasing day by day at an exponential rate. Most of the businesses are done through internet like online banking, communication, exams etc. To provide this functioning security is also needed. Internet security plays an important role in these functioning. Even though a fistful of knowledge about tools and techniques to protect networks is available, marshaled information is still eluded. Many researchers have discussed about Internet security attacks. This situation hampers the effectiveness of security[14]. This paper thus is an attempt to categorize the Internet security attack in order to comprehend the attacks more gently. The paper presented in pattern to permits internet security attacks lie under different classes of security attacks that help the developer to analysis or to avoid the causes of occurrence attack in easiest and simplest way. Thus, the purpose of this paper is to represent an idea about Classification of Internet Security Attacks. We have classified security attack into two main types: Passive attack and Active attack and also categorize it further. The various techniques used in Interception which come under passive attack viz. Release of Message Contents, Traffic analysis, Sniffing and Key loggers. Active attack is categorized into three parts i.e. Interruption, Fabrication, and Modification. The techniques that can be used under the heading interruption are DoS, DDoS, DRDoS and SQL injection attack. Replay attack and masquerading are techniques that lie under fabrication. The technique for Modification attack is man in the middle attack.

We also discuss the principles of security which help us to determine the security threats and possible security to tackle them. The principles of security are availability, accuracy, access control, confidentiality, integrity, identification &

authentication and non-repudiation. Each principle has its own importance.

Thus internet security threats will continue to be an issue as long as information is shared across the world using internet. This paper offers a few security solutions for Internet security. The efforts to devise more security techniques will continue in future to further improve the efficiency of e-commerce and communications

2. PRINCIPLE OF SECURITY

The main principles of security are [14]:

2.1 Accuracy

Information is accurate when it is free from flaws and it has the value that the user expects.

2.2 Availability

Computer resources accessible for and to authorized persons whenever required. Resources are always available for authorized person to access the data.

2.3 Authentication & identification

Authentication of information show's that the given information is genuine or original rather than fabrication. Information is authenticated when it is originally created, placed, or transferred .The data's receiver must be able to determine its origin.

2.4 Access Control

In this principle, administrator provide controls to other who should be able to access "what". For example user A can only read the database, but B can read as well as update the database.

2.5 Confidentiality

Any unauthorized person must not be able to access other's data or other computing assets

2.6 Integrity

It is related with the accuracy of data. Only authorized persons are enabled to create, edit, and delete data as per agreed terms and conditions.

2.7 Non repudiation

This principle does not allow the sender or a message to refute the claim of not sending that message. It provides protection against denies by one of the entities involved in a

communication of having participation in part of communication.

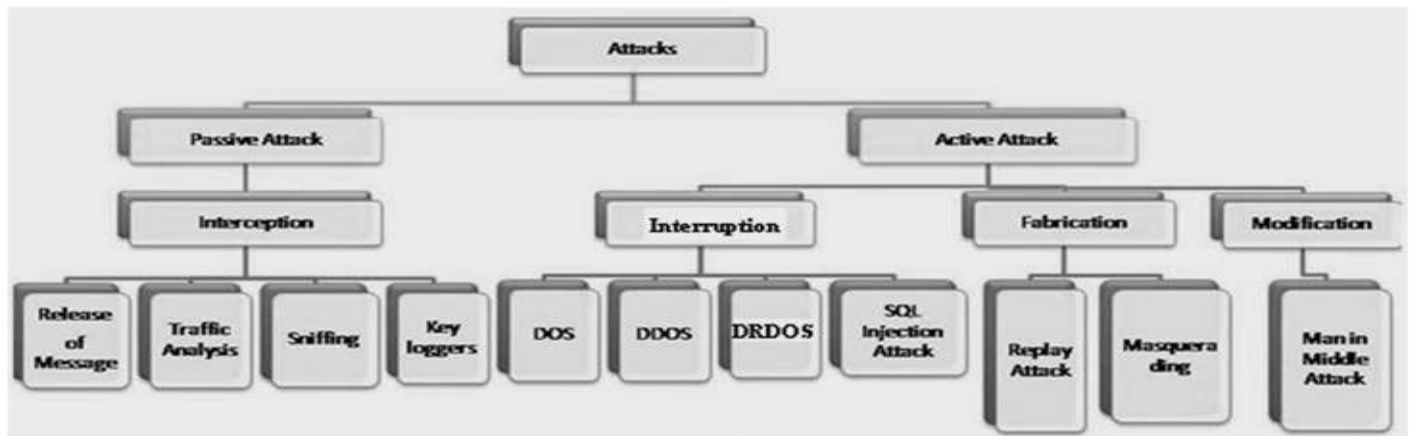


Fig: Classification of Internet Security Attacks

3. Classification of Internet Security Attack

3.1 Passive Attack:

Passive attacks are those, where the attacker aims to obtain the information. They do not wish to modify the content of original message. It is very difficult to detect as it does not alter the data. Releases of message, traffic analysis, sniffing and key loggers are some techniques of passive attacks [1][2].

3.1.1 Interception

Interception is a type of attack that is done without the permission or knowledge of the users. It breaks the rules of confidentiality in the principle of security. In simple words, we can say interception causes loss of message confidentiality. It is a type of passive attack. It is further categorize into two sub types i.e. Traffic analysis and Release of message contents. It is of four types:-

3.1.1.1 Release of message

When you send a message to you friend, you want that only that person can read the message. Using certain security mechanism, we can prevent release of message contents. For example we can encode the message using algorithm.

3.1.1.2 Traffic analysis

If many message are passes through a single channel then user get confused can give some information to attacker as it think that message is come from the his party.

3.1.1.3 Sniffing

Sniffing is a method to sniff the transferred data that was send by the sender. It just tries to find out what type of message or data is transferred by sender without the permission of sender.

3.1.1.4 Keyloggers

It is a program that runs in the background, recording all the keystrokes. Once keystrokes are logged, they are hidden in the machine for later retrieval, or shipped raw to the attacker. The

attacker then peruses them carefully in the hopes of either finding passwords, or possibly other useful information that could be used to compromise the system or be used in a social engineering attack. For example, a keylogger will reveal the contents of all e-mail composed by the user. Keylogger is commonly included in the rootkits.[11]

3.2 Active Attack

Active attacks are attacks which make some modification in the original message or creation of some false message. These attacks are very complex and cannot prevent easily. It can further categorize into 3 types: Interruption, Fabrication, and Modification. Under these categorize Denial of service (DoS), DDoS, DRDoS, SQL Injection, Replay attack, Masquerading, Man in Middle Attacks are some common attacks[1][2].

3.2.1 Interruption

Interruption attacks are active attack. In this attack an authorized entity pretends to be another entity. For example there are three users A, B & C. User A might be pose as user C and send a message to user B. User B believe that message came from user C. Interruption puts the availability of resource in danger. It is classified into four types

3.2.1.1 Denial of Service (DoS)

When a system receiving the requests becomes busy trying to establish a return communication path with the initiator (which may or may not be using a valid IP address) and remains in a wait condition due to which legitimate users are denied to access.[03] [04][08].

3.2.1.2 Distributed Denial of Services (DDoS)

On the internet, a distributed denial-of-service (DDoS) attack is one in which a multitude of compromised systems attack a single target, thereby causing denial of service for users of the targeted system. The flood of incoming messages to the target

system essentially forces it to shut down, thereby denying service to the system to legitimate users [3][4][8].

3.2.1.3 Distributed DoS with Reflectors (DRDoS)

It consists a reflector that helps the attacker to execute a more effective and secure attack. It results in increase of damage and decrease the risk of being traced back [4][8].

3.2.1.4 SQL Injection Attack

SQL injection is a security vulnerability that occurs in the database layers of an application. It is the act of passing SQL code into interactive web applications that employ in database services [12].

3.2.2 Fabrication

In this attack users use some accessing service, which they are not eligible for. It is possible in the absence of proper authentication mechanisms [2][13][14]. Under these two attack technique used

3.2.2.1 Replay Attack

A replay attack is a form of active attack in which a valid data transmission is maliciously repeated or delayed. A attacker captures the authorized data and resends them to his personal use. For example User A wants to transfer some amount to User C's Bank account. Both User A & C have account with Bank B. User A send a electronic message to Bank B, requesting a fund transfer. User C could capture this message, and send a second copy to Bank B but Bank B could not have idea that this is an unauthorized message. Thus User C would get benefit of fund transfer twice. A replay attack can be prevented using strong digital signatures that include time stamps and inclusion of unique information from the previous transaction such as the value of a constantly incremented sequence number.

3.2.2.2 Masquerading

Masquerading attack is a type of attack in which one system assumes the identity of another. It's a technique used by attacker to pretend himself as an authorized person in order to gain access of confidential information in illegal manner [13].

3.2.3 Modification

Modification causes losses of integrity principle. For example a person did an online transaction of Rs. 100. But the attacker hack this and modify it to Rs.1000. This is a case of integrity. Under this attack technique is man of the middle attack [2][7][14].

3.2.3.1 Man of the middle attack

It is abbreviated as MITM. It is an active internet attack that attempts to intercept, read and alter the information hovering between the user of a public network and any requested website. The attacker uses the illegally gained information for identity theft and other types of fraud [7].

4. CONCLUSION

Internet security threats will continue to be an issue as long as information is shared across the world using Internet. This paper offers a classification and few security solutions for Internet security attack. The efforts to devise more security techniques will continue in future to further improve the efficiency of e-commerce and communications.

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